

- (b) Discuss the role of transactions in a database and its properties?

(Or)

7. (a) Discuss the role of views in enhancing the security of a database.
- (b) Discuss the Two-Phase Locking (2PL) protocol in detail. Include its advantages, disadvantages, and application scenarios.

(Or)

8. (a) Discuss the challenges associated with maintaining functional dependencies during the normalization process.
- (b) Explain the concept of lossless decomposition and its importance in dependency preservation.

(Or)

9. (a) Define Specialization and Generalization with example.
- (b) Explain Database recovery techniques.

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Roll No.

Paper Code – BCA501(N)

BCA 3rd Year (Semester-V) EXAMINATION, 2023-24

INTRODUCTION TO DATABASE

MANAGEMENT SYSTEM

PAPER-I

Time : Two Hours]

[Maximum Marks : 75

Note- This paper consists of three Section A, B and C.

Carefully read the instructions of each Section in solving the question paper. Candidates have to write their answers in the given answer-copy only. No separate answer-copy (B Copy) will be provided.

Section-A

(Short Answer Type Questions)

Note- All questions are compulsory. Answer the following questions as short answer type questions. Each question carries 5 marks.

1. (A) What is the purpose of a Primary Key in a database?
- (B) Briefly explain the concept of normalization?

- (C) Define the term "Foreign Key"?
- (D) What is SQL and what does it stand for?
- (E) Explain the difference between a table and a view in a database?
- (F) Define the term "ACID Properties" in the context of databases?
- (G) What are the main difference between a file processing system and DBMS?
- (H) What is difference between logical data independence and physical data independence?
- (I) Differentiate between weak and strong entity set?

Section-B

(Long Answer Type Questions)

Note- This section contains four questions from which **one** question is to be answered as long question. Each question carries 15 marks.

- 2. (a) Discuss the advantages and disadvantages of using a relational database model.
- (b) Briefly explain the concept of data integrity in a database.

(Or)

- 3. (a) What is Data Model? Also explain their types?
- (b) Discuss the three level architecture of database system?

(Or)

- 4. (a) What is the purpose of the GROUP BY clause in SQL?
- (b) Construct an E-R Diagram for University Computer application department admission process?

(Or)

- 5. (a) Explain sequential file organization?
- (b) What is extendible hashing?

Section-C

(Long Answer Type Questions)

Note- This section contains four questions from which **one** question is to be answered as long question. Each question carries 15 marks.

- 6. (a) Describe the three normal forms in database normalization.